

Practice Quiz: The Prediction Machine

Part A: Pick the Best Answer

1. What is your Brain always doing? A) Sleeping B) Predicting the future C) Counting backward from 100 D) Singing opera

Answer: B! Great job! Your brain is a prediction machine that is always asking "What will happen next?"

2. If you see a dark cloud, what is a good prediction? A) It will rain B) It will rain candy C) The sun will explode D) A dinosaur appears

Answer: A! That's right! Your brain remembers that dark clouds usually mean rain. That is a smart prediction!

3. What happens when your prediction is WRONG? A) You vanish B) You feel Surprised C) You turn green D) Nothing

Answer: B! Exactly! Surprise is your brain saying "Whoa! I did not expect that!" And that surprise helps you learn.

4. Why do we need memories? A) To help us guess what will happen next B) To fill up space C) They are heavy D) To forget things

Answer: A! Nice work! Your brain uses memories like a map to predict the future. The more you remember, the better your guesses!

5. If you let go of a balloon, what do you predict? A) It will fall down (if full of air) B) It will float up (if full of helium) C) It will turn into a bird D) Both A or B

Answer: D! Smart thinking! It depends on what is inside the balloon. Your brain uses what it knows to make the right prediction.

6. What is a "Pattern"? A) Something that repeats (A-B-A-B) B) A messy room C) A single shoe D) A noise

Answer: A! That's right! A pattern is something that repeats in a way you can predict, like Clap-Stomp-Clap-Stomp.

7. Surprise helps us: A) Sleep B) Learn C) Eat D) Jump

Answer: B! Exactly! When you are surprised, it means your prediction was wrong, and now your brain can update its map. That is learning!

8. Your brain builds a map called a "Generative Model." What does that map remember?

A) Only your name B) Everything that happened before, like lemons being sour C) Nothing at all D) Only things that are blue

Answer: B! Good thinking! Your brain remembers all kinds of things: what foods taste like, where your house is, and that fire is hot.

9. If you pull 1 block from a bag with 3 blue blocks and 1 red block, what color do you predict? A) Red (because it is special) B) Blue (because there are more blue ones) C) Green (because green is pretty) D) Purple (because it is your favorite)

Answer: B! Nice! Since there are more blue blocks, your brain predicts blue is more likely. That is smart thinking!

10. When your prediction matches what really happens, how does your brain feel? A) Confused B) Happy and calm because the world makes sense C) Angry D) Sleepy

Answer: B! That's right! When you guess correctly, your brain feels good because everything makes sense. No surprises needed!

Part B: Tell Us What You Think

1. What comes next: Square, Circle, Square, Circle, _____?
2. If you smell cookies baking, what do you predict you will taste soon?
3. Draw a picture of something you predict will happen tomorrow.
4. Think about the Magic Bag activity. If you put 5 red blocks and 1 blue block in a bag, which color are you most likely to pull out? Why?

5. Can you think of a time when your brain predicted something wrong but you learned from it?